MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION …….. 2025

THIRD SEMESTER

MG3DSEECO200 - Introduction to Data Analysis for Economics

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A Multiple Choice Questions

Answer all questions; Each question carries 1 marks

1. In Excel, the formula =AVERAGE(A1:A5) returns: [K] / [CO1]

   (a) The maximum value in the range (b) The median value

   (c) The sum of values (d) The arithmetic mean

2. Which function will count only numeric entries in a dataset? [K] / [CO1]

   (a) COUNT() (b) COUNTA()

   (c) COUNTIF() (d) SUM()

3. Which chart is most appropriate to visualize the proportion of different categories in a whole? [U] / [CO2]

   (a) Bar chart (b) Line chart

   (c) Pie chart (d) Scatter plot

4. The function used to combine two text values in Excel is: [K] / [CO1]

   (a) JOIN() (b) ADDTEXT()

   (c) CONCATENATE() (d) TEXTMERGE()

5. Which Excel function returns the number of characters in a string? [K] / [CO1]

   (a) LEN() (b) COUNT()

(c) CHAR() (d) FIND()

6. A data model that expresses relationships using algebraic formulas is known as: [U] / [CO4]

(a) Pie model (b) Logical model

(c) Algebraic model (d) Forecasting model

7. Which function returns the standard deviation of a dataset in Excel? [K] / [CO3]

(a)MEAN() (b) SD()

(c) STDEV() (d) VAR()

8. Pearson's correlation coefficient lies between: [U] / [CO3]

(a) -2 and +2 (b) -1 and +1

(c) 0 and 1 (d) -10 and +10

9. Which Excel function is most suitable to extract the first three characters from a cell? [K] / [CO1]

(a) LEFT() (b) MID()

(c) RIGHT() (d) CHAR()

10. The INTERCEPT() function in Excel is used in: [U] / [CO4]

(a) Interest calculation (b) Regression analysis

(c) Filtering datasets (d) Text splitting

11. A scatterplot is most useful for examining: [U] / [CO3]

(a) Relationship between two variables (b) Frequencies of categories

(c) Differences in medians (d) Compound interest

12. The ROUND() function in Excel is used to: [U] / [CO1]

(a) Add two numbers (b) Convert text to numbers

(c) Remove duplicates (d) Adjust numbers to a specified number of digits

13. A pivot table is most useful when you want to: [U] / [CO3]

(a) Format data (b) Perform correlation

(c) Summarize large datasets dynamically (d) Validate formulas

14. The relationship between two variables in regression is often shown by: [U] / [CO4]

(a) A pie chart (b) A regression line

(c) A column chart (d) A histogram

15. Which of the following measures relative variation? [U] / [CO3]

(a) Variance (b) Standard deviation

(c) Coefficient of Variation (d) Median

PART B

Short Answer Questions

Answer any 10 questions.

Each questions carries 2 marks.

16. Data analysis. [U] / [CO1]

17. MS Excel. [U] / [CO1]

18. Data sorting. [K]/ [CO2]

19. Measures of central tendency. [U]/ [CO3]

20. Variance and Standard deviation. [U]/ [CO3]

21. Annuities. [K]/ [CO4]

22. Compound interest. [U]/ [CO4]

23. Pivot tables and Pivot Chart. [U]/ [CO2]

24. Coefficient of variation. [U]/ [CO3]

25. Functions in Economics. [U]/ [CO2]

26. Correlation. [K]/ [CO3]

27. Internal Rate of Return. [K]/ [CO4]

28. Operators. [K] / [CO1]

29. Scatter plot. [K]/ [CO2]

PART C

Short Essay Questions

Answer any 3 questions.

Each questions carries 5 marks.

30. Following table gives monthly sales data for a company. [A] / [CO1]

A	B	C
1	**Month**	**Sales ($)**
2	January	5,000
3	February	4,800
4	March	5,500
5	April	6,000
6	May	5,200
7	June	6,300

Based on this data, write the functions to find the highest and lowest sales, sum, average and find the values in round figure.

31. What is a LOOKUP function? How VLOOKUP and HLOOKUP are different from each other? [U]/ [CO2]

32. Based on the scatterplot given below, answer the following questions. [An]/ [CO4]



a) Create a table for price and quantity demanded from the above graph.
b) What does the negative slope indicate in economic terms?
c) If quantity demanded (Q_d) = 123.6 – 5.7* Price, then what is the quantity demanded when price is 10?
d) Use the excel functions to find the slope and intercept from the table you have created.
e) Identify the analysis used in the graph and write a short note on it.

33. From the given household dataset; find median, range, variance and standard deviation. Mention the associated statistical functions used in Excel. [A]/ [CO3]

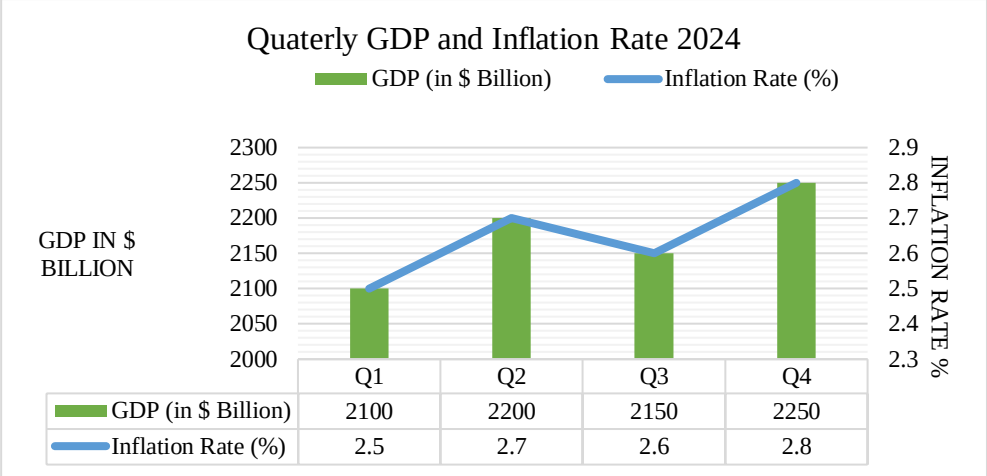
Household	**Food Spending ($)**
H1	320
H2	290
H3	310
H4	280
H5	350

34. "MS Excel is a powerful tool in data analysis". Substantiate your answer. [U] / [CO1]

35. What is correlation coefficient? Find the correlational coefficient of the below data and identify the nature of relationship. Write the Excel function. [A] / [CO3]

Student	**Study hours**	**Exam score obtained**
A	2	55
B	4	65
C	6	75
D	8	85
E	10	95

36. The following chart gives GDP and inflation rate trends for four quarters of 2024. Based on this chart, answer the following questions. [An] / [CO2]



a) Which quarter recorded the highest inflation rate?

b) Which quarter recorded the lowest GDP?

c) What is the GDP growth from Q1 to Q4 in percentage?

d) What is the mean inflation rate?

e) What is your inference about the relationship between GDP and inflation rate?